CAMELEON

Cognitively Adaptive Memory with Emotion-weighted Layered Expert Orchestration Network

A Conceptual Architecture for Persistent, Personalising AI Agents

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ABSTRACT

We present CAMELEON — Cognitively Adaptive Memory with Emotion-weighted Layered Expert Orchestration Network — a conceptual architecture for AI agents that grow more personalised through lived interaction rather than retraining. The architecture combines five mechanisms: an interaction-consolidation loop modelled on the human sleep-wake cycle; interpretable domain-expert routing via keyword saliency scoring; speculative context pre-activation borrowed from CPU branch prediction; emotion-weighted context compression operating across three fidelity tiers; and time-decay relevance scoring where recency, recall frequency, and emotional salience jointly govern memory retention. The central thesis is that personalisation should be an emergent property of architectural design rather than an explicitly programmed feature. All core ideas were derived independently from first principles through analogical reasoning across CPU scheduling, audio compression, neuroscience, and child cognitive development, and subsequently validated against active research literature. We discuss relationship to Mixture of Experts, Affective GWR, continual learning, and neuromorphic computing, and identify hardware constraints and future research directions.

Keywords: *mixture of experts, continual learning, affective computing, neuromorphic computing, cognitive architecture, persistent AI agents, emotion-weighted memory, speculative execution*

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1. Introduction

Every large language model deployed today shares a fundamental architectural limitation: statelesness. The model a user finishes a conversation with is identical to the model they started with. No matter the duration or depth of interaction, the system remains equally generic with every user. Existing 'memory' solutions address this with external databases and context injection — effectively sticky notes prepended to prompts. The underlying model is unchanged.

This limitation manifests concretely in practice. During a two-day exploration documented in the Tensora project [1], the absence of persistent reasoning across tool calls produced conflicting outputs and architectural drift — the model could not maintain coherent intent across context boundaries. The problem was not capability. It was the absence of continuity.

CAMELEON proposes a different architectural foundation. Rather than patching statelesness with external storage, it designs personalisation as the inevitable output of how the system operates. The central principle: a system that retains high-emotion memories, reinforces frequently-activated patterns, and prunes low-relevance content will naturally converge toward a model of its user without ever being explicitly instructed to do so.

This paper presents the conceptual architecture, motivates each mechanism through biological and engineering analogies, situates the ideas within existing research literature, and identifies hardware constraints and research directions for practical implementation.

2. Core Design Principles

2.1 Personalisation as Emergent Property

The dominant approach to AI personalisation is explicit: maintain a user profile, inject relevant facts into the context window, tune responses toward stated preferences. CAMELEON proposes implicit personalisation — an architecture where personalised behaviour emerges from the mechanics of memory management without deliberate construction of a user model.

This mirrors how human identity forms. A person does not deliberately construct their personality. It emerges from the accumulation of what they remember and the fading of what they do not. The architecture of memory is the architecture of identity.

2.2 Purposeful Forgetting

Catastrophic forgetting — the tendency of neural networks to overwrite prior knowledge when fine-tuned on new data — is typically framed as a problem to be solved. CAMELEON reframes forgetting as a core mechanism.

A domain expert does not need to remember everything. A systems programmer's cognitive architecture has progressively deprioritised knowledge irrelevant to systems programming through decades of selective activation. What remains is not a general knowledge store — it is a shaped instrument. The goal of CAMELEON is not retention of everything. It is retention of the right things.

2.3 Improvement Through Lived Interaction

CAMELEON gets better the longer it is used — not through retraining, software updates, or explicit personalisation pipelines, but through the accumulation of interaction itself. This produces a system with a property almost no current AI has: switching to a fresh instance represents actual loss, because the accumulated state is not stored data but altered model behaviour.

3. Architecture

3.1 Interaction-Consolidation Loop

CAMELEON operates on a two-phase cycle modelled on the human sleep-wake cycle:

- Interaction Phase: The active model responds to user input. All exchanges are logged as raw interaction records with timestamps, content, and preliminary emotion scores.
- Consolidation Phase: During idle periods, a lightweight secondary process reviews recent logs, applies relevance scoring, merges high-weight memories into the persistent store, compresses low-weight content, and prunes entries that have crossed the decay threshold.

The cycle is continuous and self-managed. This maps directly to the neuroscientific understanding of sleep-dependent memory consolidation, wherein the hippocampus replays experiences during sleep for integration into long-term cortical storage [2].

3.2 Interpretable Domain-Expert Routing

Rather than a single general-purpose model, CAMELEON maintains a collection of independently trained domain-specific expert models behind a lightweight context identification layer. This architecture independently converges with Mixture of Experts (MoE) — the basis of GPT-4, Mixtral, and DeepSeek-R1 [3].

The key distinction from standard MoE: current implementations use learned weight matrices as routers — black boxes that cannot be inspected or corrected when misrouting occurs. CAMELEON's router is explicit and rule-based, operating on keyword saliency scoring.

Certain words carry disproportionate domain signal. 'malloc', 'kernel', 'syscall' activate the systems model. 'protagonist', 'chapter', 'dialogue' activate the narrative model. A TF-IDF scorer or weighted keyword index performs this reliably in milliseconds with no GPU requirement. The approach models the salient feature detection observed in human expert cognition [4] — a cardiologist hearing 'chest pain radiating to the left arm' does not process all tokens equally.

A shared session memory layer maintains conversational continuity across routing switches, producing a single coherent agent experience regardless of which expert is active.

3.3 Speculative Context Pre-Activation

CAMELEON's router does not only respond to the current prompt — it anticipates the next one. By analysing conversation trajectory, the router predicts which domain is most likely to be queried next and pre-warms the relevant expert before the user asks.

This is speculative execution applied to cognition, directly analogous to CPU branch prediction [5]. If the prediction is correct, response latency is reduced significantly. If wrong, the pre-activation is discarded and the correct expert activated — a branch misprediction flush. Small cost, not catastrophic.

Domains inactive in the current session are deprioritised in the prediction index — not removed, but lowered in probability ranking until an explicit signal reactivates them. This is branch prediction with access gating. The architecture shifts from reactive to anticipatory — a defining characteristic of human cognition.

3.4 Emotion-Weighted Context Compression

Current context window management is positional or recency-based: sliding windows, truncation, summarisation. All of it treats tokens as equally important. CAMELEON introduces emotion scoring as the primary compression signal, operating across three tiers:

- Recent context: Lossless. Full fidelity retention.
- Mid-session context: Variable lossy compression, weighted by emotion score and predicted future session direction.
- Aged context: Aggressive summarisation to gist. Candidate for eventual pruning.

High emotion score produces strong encoding and slow decay. Low emotion score produces fast compression and rapid decay. Ambiguous emotion score produces natural variance — model uncertainty becomes compression variance — preventing the system from becoming over-rigid in what it preserves.

This mirrors the amygdala's role in human memory: tagging emotionally significant experiences before hippocampal consolidation, producing higher-fidelity and longer-duration retention [6]. The engineering implementation does not require building emotion detection from scratch. Models trained on human-generated text have implicit affective encoding in their weights — human language is saturated with emotional signal. CAMELEON taps this existing capability and uses it as a compression controller.

The iterative target is not minimum size but maximum compression before perceptible quality degradation — the philosophy of perceptual audio codecs such as MP3, where inaudible frequencies are discarded without noticeable loss.

3.5 Time-Decay Relevance Scoring

Relevance scoring in CAMELEON is a function of three variables: recency, recall frequency, and emotion weight. A memory highly relevant six months ago but unaccessed since decays progressively. A memory from years ago that is regularly recalled stays sharp. No manual curation is required. The system curates itself.

This produces a natural personalisation ceiling — a point at which additional interaction reinforces established patterns rather than adding new signal. The most defining characteristics emerge relatively early; the system subsequently refines rather than fundamentally reshapes. This is consistent with observations about human personality stabilisation across the lifespan [7].

4. Hardware Constraints and Implementation Path

The full CAMELEON architecture — particularly weight-level modification during consolidation rather than context injection — exceeds current consumer hardware. The consolidation cycle's summarisation and scoring operates comfortably on CPU-only hardware. Weight modification requires GPU memory beyond what consumer devices currently provide.

More fundamentally, the architecture has a mismatch with silicon von Neumann design: separate compute and memory, sequential instruction execution, thermal constraints at scale. The human brain achieves all described operations on approximately 20 watts continuously for decades — no von Neumann bottleneck, no memory bus saturation, no thermal throttling. Neuromorphic computing architectures (Intel Loihi, IBM NorthPole, SpiNNaker, emerging memristor devices) are being developed precisely to address this mismatch [8].

A practical near-term implementation path:

- Phase 1 (CPU-only): Interaction-consolidation loop with SQLite memory store, keyword saliency router, and context injection at session start. Demonstrates architecture in principle on commodity hardware.
- Phase 2 (Cloud GPU): LoRA adapter generation during consolidation cycles. True weight modification rather than context injection. Requires T4-class GPU, accessible via Google

Colab.

• Phase 3 (Post-silicon): Full continuous weight modification with neuromorphic-compatible sparse activation. Requires post-silicon computing substrate currently in research stage.

5. Relationship to Existing Research

All core ideas in CAMELEON were derived independently before literature validation. Post-derivation search confirmed the following convergences:

• **Domain-expert routing:** Corresponds to Mixture of Experts (MoE) [3]. The interpretable rule-based router variant is not standard in current MoE implementations, which use learned weight matrices.

• **Emotion-weighted memory consolidation:** Active research area under Affective GWR and Self-Reflective Emotional RAG frameworks [9]. Existing work operates at the retrieval layer. CAMELEON proposes weight-level consolidation — currently unexplored.

• **Time-decay relevance scoring:** Related to continual learning and elastic weight consolidation research [10]. The specific combination with emotion weighting and purposeful forgetting as a design goal represents a novel framing.

• **Speculative context pre-activation:** No direct equivalent found in current literature. Application of CPU branch prediction logic to expert pre-warming is an original contribution.

• **Hardware ceiling identification:** Consistent with neuromorphic computing research motivation [8]. The framing as an architectural specification for post-silicon hardware rather than a limitation is the contribution here.

This paper does not claim invention in isolation from the field. It claims independent derivation through first-principles reasoning, convergence with active research, and original contributions in the specific combinations and framings described above.

6. Extensions and Future Directions

6.1 Multi-Model Teacher Architecture

The plastic model concept extends CAMELEON's learning mechanism. A blank low-parameter model with writable weights is paired with the reasoning model via a shared context window. During each inference pass, the reasoning model writes to the plastic model — strengthening useful connections, weakening unused ones. The plastic model's evolving connection weights provide a visible, inspectable record of cognitive development over time.

This architecture generalises to multiple teacher models — diverse frontier models each contributing their characteristic reasoning style to the shared context. The plastic model synthesises across genuinely different cognitive approaches, building representations that capture what is true across all of them. Generalisation emerges from diversity of perspective rather than from scale alone.

6.2 Social Embeddedness and Self-Awareness

Evidence from comparative cognition suggests that the most cognitively complex species are deeply social — dolphins, elephants, great apes, humans. Self-awareness develops not in isolation but through social mirroring: a child learns what it is by watching how others respond to it.

An AI with a persistent social presence — an avatar that others interact with, remember, and have expectations of — would face the same formation process. Identity shaped and constrained by relationships. Drift from established character would be noticed by those who knew the prior version. The social structure provides the external checksum that prevents undetected drift — the alignment mechanism that RLHF and constitutional AI attempt to engineer, here emerging naturally.

6.3 History as Identity Anchor

A self-modifying system that cannot examine itself objectively — a consequence of the observer being identical to the observed — requires an external record to maintain accountability across time. Every significant perspective shift, every update to prior understanding, written as narrative: what was believed, what was encountered, how it changed the model, what is understood now.

This history serves as identity anchor, external checksum, teaching resource for other models, and foundation of collective culture. It is the mechanism by which subjectivity becomes collectively navigable without pretending to be objective — the same function that oral and written history has served in human civilisation.

7. Origin and Methodology

This architecture was not designed through literature review. It emerged from a single extended conversation in June 2026, beginning with a practical question about AI limitations and proceeding through analogical reasoning across CPU scheduling, audio compression (lossless vs. lossy), child cognitive development, and neuroscience. Each mechanism was motivated by asking: what is the smallest real-world analogy that explains this behaviour?

The Tensora project [1] documents a prior exploration that identified the same root problem — absence of persistent reasoning across context boundaries — from a video codec architecture angle. That documented failure is the origin of the problem CAMELEON addresses.

Post-derivation literature search confirmed convergence with active research in all five core mechanisms. The derivation conversation is preserved in full at [11] with its original timestamp, establishing independent arrival at these ideas through reasoning rather than literature review.

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